MCU Relay Bypass Module (True Bypass)

MCU Firmware: https://github.com/matt-garman/mcu-bypass-firmware

Reference	Value	Footprint	Notes
R13	100R	DIN0204 / 3.6 x 1.6 x 5.08 mm	Forms PSU filter with C1
R29	1k	DIN0204 / 3.6 x 1.6 x 5.08 mm	Footswitch RC filter with C29
CLR1	4k7	DIN0204 / 3.6 x 1.6 x 5.08 mm	Current-limiting resistor, adjust to taste
R22	10k	DIN0204 / 3.6 x 1.6 x 5.08 mm	MCU footswitch pin pullup resistor
R23	10k	DIN0204 / 3.6 x 1.6 x 5.08 mm	MCU reset pin pullup resistor
R24	100k	DIN0204 / 3.6 x 1.6 x 5.08 mm	LED mosfet gate pulldown resistor
R27	100k	DIN0204 / 3.6 x 1.6 x 5.08 mm	Relay reset mosfet gate pulldown resistor
R30	100k	DIN0204 / 3.6 x 1.6 x 5.08 mm	Relay set mosfet gate pulldown resistor
C29	22nF	Box film cap 7.2 x 2.5 mm	Footswitch RC filter with R29
C28	100nF	Ceramic X7R or box film cap 7.2 x 2.5 mm	MCU supply decoupling
C30	470nF	Ceramic X7R or box film cap 7.2 x 2.5 mm	5v LDO supply decoupling/filtering
C8	10uF	Ceramic X7R 7.2 x 2.5 mm	5v bulk supply decoupling
C1	>=10uF	Ceramic X7R or Electrolytic 5mm diameter (low profile)	Forms PSU filter with R13; bigger is better (as space allows)
D1	1N5817	DO-35	Relay coil flyback protection diode
D2	1N5817	DO-35	Reverse polarity protection
D3	1N5817	DO-35	Relay coil flyback protection diode
Q1	2N7000	TO-92	other N-MOS ok, but mind pinout
Q2	2N7000	TO-92	other N-MOS ok, but mind pinout
Q3	2N7000	TO-92	other N-MOS ok, but mind pinout
K1	TQ2-2L-5V	Panasonic TQ2-2L-5V or -4.5V relay	
U1	LP2950-5.0_TO92	TO-92	5v LDO; 78L05 ok but mind pinout
U9	ATtiny13A	DIP-8	Can also use ATtiny45, ATtiny85